Task-Dependent Functional Connectivity with the Cerebellum: Confounded by Nyquist Ghosting?

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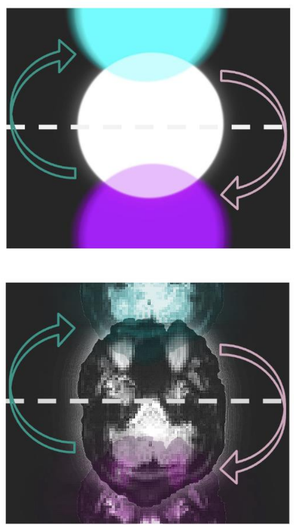
1. Introduction

- Recent studies have implicated the cerebellum in functions beyond its traditional role in motor coordination: e.g., social cognition (Van Overwalle et al., 2020), decision making (Schmahmann et al., 2019), memory (Brissenden, 2019)
- These may be confounded by artifacts tied to eye movements (i.e., Nyquist ghosting)

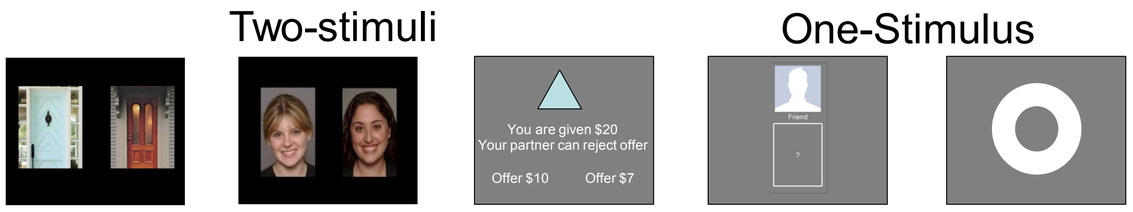


This study addresses three main questions:

1. Are there correlations between signal changes in the eyes and cerebellum?
2. Do these correlations depend on the tasks?
3. Is eye~CB signal related to the amount of ghosting?

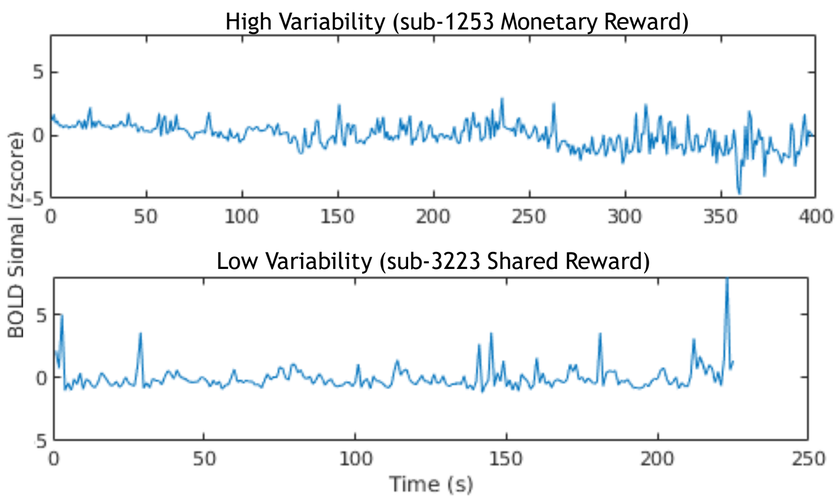
2. FMRI-based tasks

Data from five fMRI-based tasks (N=52) were sorted into groups based on visual stimuli: tasks with two stimuli (i.e., greater hypothesized eye movements) vs. tasks with one stimulus (i.e., lesser eye movements):



Monetary Reward | Social Reward | Ultimatum Game | Shared Reward | Monetary Incentive Delay

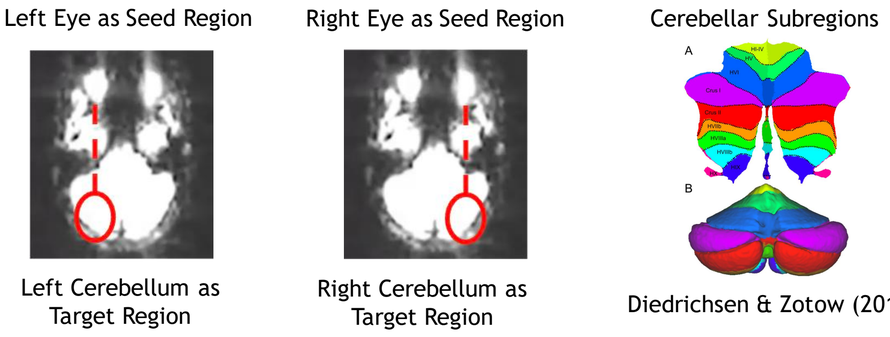
3. Task-dependent variability in eye movements



Visualization of differences in variability of eye movement across the timecourse of functional runs between tasks and across participants, potentially arising from differences in task design and stimuli

4. Neuroimaging methods & analyses

Left Eye as Seed Region | Right Eye as Seed Region | Cerebellar Subregions

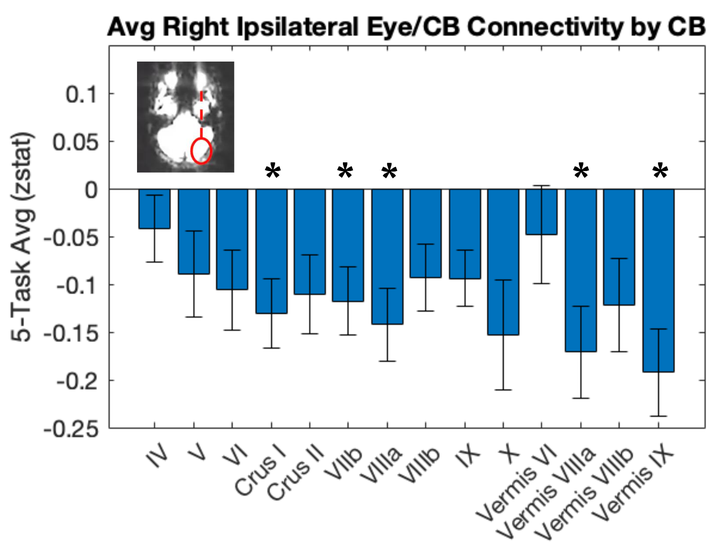


Left Cerebellum as Target Region | Right Cerebellum as Target Region | Diedrichsen & Zotow (2015)

Functional connectivity with each eye was measured in two generalized psychophysiological interaction (PPI) models (one per hemisphere), with the eye as a seed region and cerebellar subregions (Diedrichsen, 2009) as targets

5. Significant connectivity between right eye and cerebellar subregions

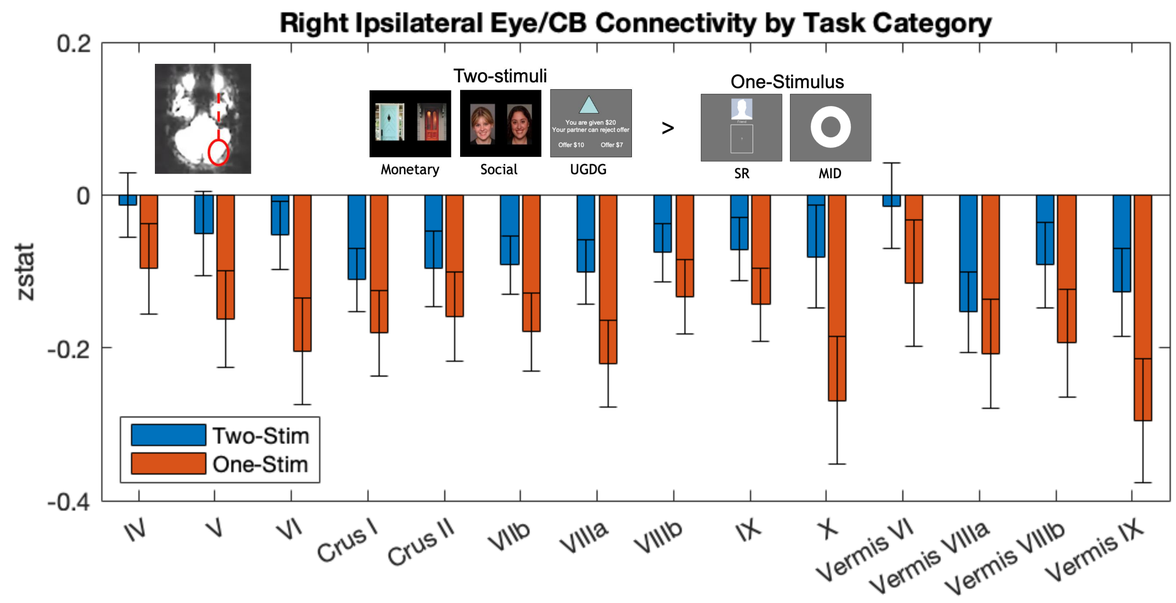
We first examined whether there is significant PPI signal between the eyes and any region of the cerebellum, taking the average signal across our five tasks:



Findings show that yes, there is significant connectivity between the right eye and right hemisphere in 5 cerebellar subregions: Crus I (t=2.82, $fwep$<.05), VIIb (t=2.87, $fwep$<.05), VIIIa (t=3.17, $fwep$<.05), Vermis VIIIa (t=2.94, $fwep$<.05), and Vermis IX (t=3.91, $fwep$<.01)

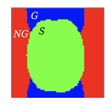
6. Assessing whether PPI signal between eye and cerebellum depends on task

Next, we predicted that eye~cerebellum connectivity would be greater for two-stimuli vs. one-stimulus tasks, indicating task-dependency



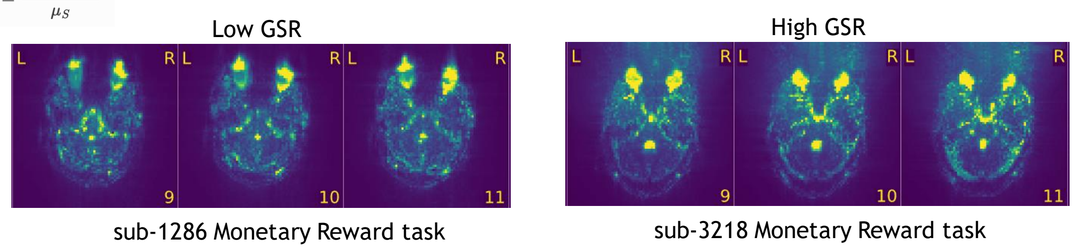
No significant differences emerged between two-stimuli vs. one-stimulus tasks

7. Is PPI signal between eye and cerebellum related to ghost-to-signal ratio (GSR)?



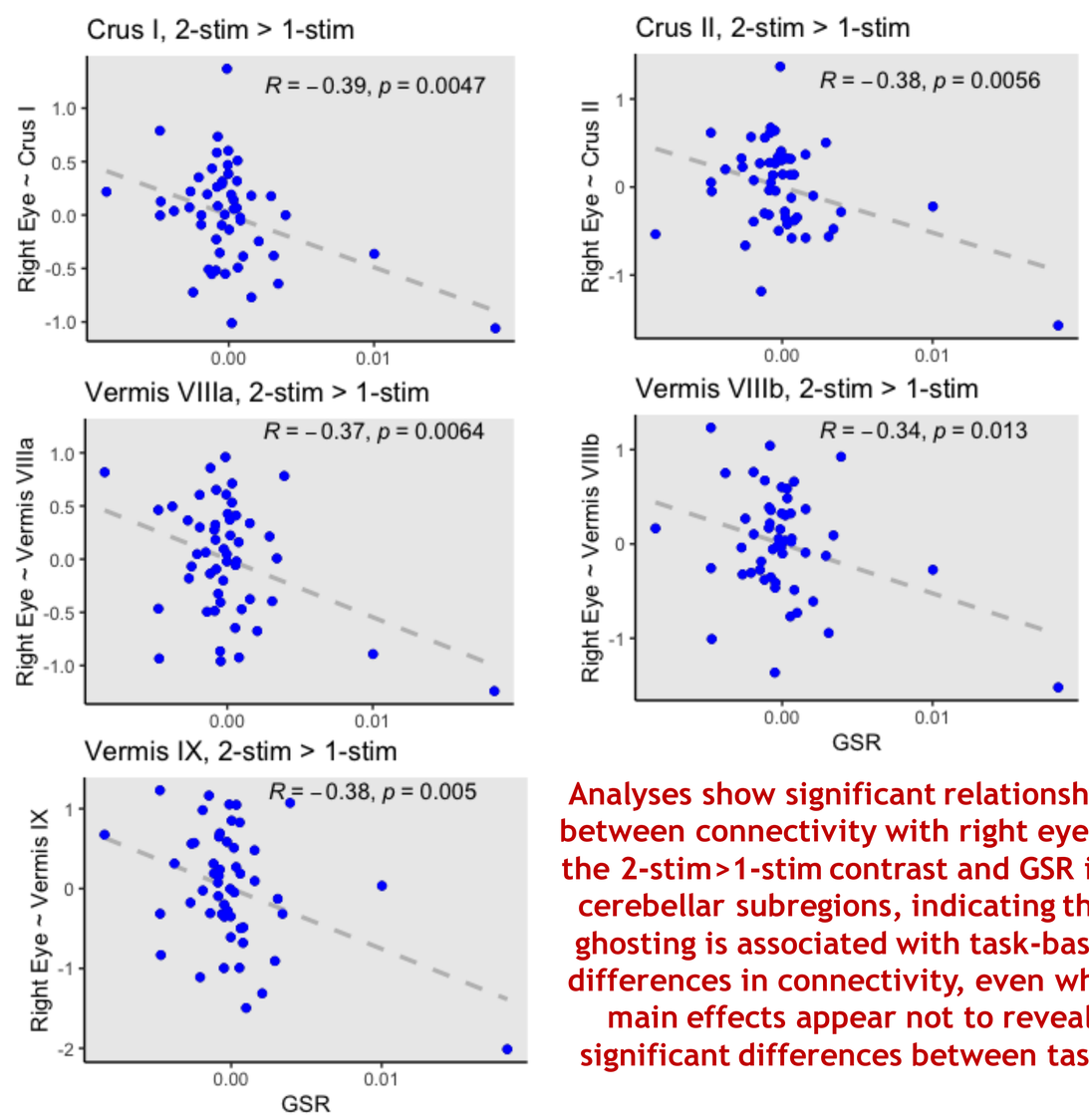
$$\mathrm{GSR} = \frac{\mu_G - \mu_{NG}}{\mu_S}$$

The magnitude of ghosting differs between participants. This can be quantified via MRIQC's ghost-to-signal ratio (GSR), a default metric that estimates the mean signal intensity in areas prone to ghosting



sub-1286 Monetary Reward task | sub-3218 Monetary Reward task

We predicted that GSR would be associated with the difference in connectivity for 2-stim vs 1-stim tasks



Analyses show significant relationships between connectivity with right eye for the 2-stim>1-stim contrast and GSR in 5 cerebellar subregions, indicating that ghosting is associated with task-based differences in connectivity, even when main effects appear not to reveal significant differences between tasks

8. Conclusions

- Findings show that signal from eyes and cerebellum is significantly correlated, and that the strength of this relationship between tasks is associated with GSR
- Importantly, while it might not look like ghosting is impacting signal when just assessing main effects between tasks, inclusion of GSR as a covariate may reveal this relationship

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